

Explainable Deep Learning for Multi-Modal Medical Imaging

Toward Trustworthy AI-Assisted Diagnosis

MedTrustXAI

MedTrustXAI - Multi-Modal Research Prototype & Implementation Report

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Abstract

This project presents MedTrustXAI, an explainable deep learning framework for multi-modal medical imaging that supports both radiology (chest X-ray) and digital pathology (H&E histopathology). The system integrates a biomedical Vision-Language Model (BMC-SmolVLM-256M) with visual attribution (Grad-CAM / input-gradient saliency), modality-specific faithfulness evaluation, WSI patch tiling for large pathology slides, and quantitative VQA/report metrics. A Gradio interface and CLI provide interactive demonstration, batch inference, and evaluation on curated test datasets.

1 Introduction

Trustworthy AI-assisted diagnosis requires models that not only predict accurately but also explain their reasoning in clinically inspectable form. Radiology and digital pathology are complementary imaging modalities: radiographs reveal macroscopic anatomy while histopathology reveals cellular and tissue architecture. MedTrustXAI unifies both under a single explainable VLM pipeline with modality-aware prompts, region maps, and metrics.

2 Implemented Modalities

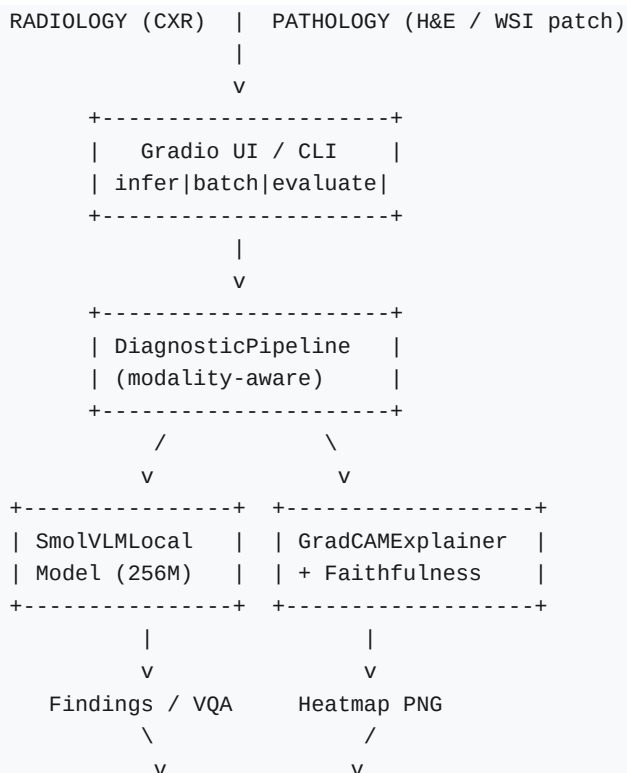
2.1 Radiology (Chest X-ray)

- 8-sample test dataset (normal, pneumonia, VQA-RAD chest)
- Anatomy-based faithfulness regions (lungs, heart, pleura, mediastinum)
- Prompt presets: full report, findings, diagnosis, VQA questions

2.2 Digital Pathology (H&E)

- 4-sample test dataset (seminoma, renal carcinoma, metastatic ductal carcinoma, hepatic amyloidosis)
- Tissue-based faithfulness regions (epithelium, stroma, nuclei, tumor, necrosis)
- WSI patch tiling: center patch or grid extraction from large slides

3 System Architecture



```
+-----+
| outputs/ JSON + PNG |
+-----+
```

4 Explainability and Trust

- Grad-CAM on CUDA; fast input-gradient saliency on Mac MPS/CPU
- Modality-specific faithfulness: anatomy (radiology) vs tissue regions (pathology)
- Follow-up chat in Gradio for clinician-style questioning
- Saved JSON artifacts for reproducibility and guide review

5 WSI Patch Tiling

Large whole-slide images are tiled into fixed-size patches (default 512x512). The center-patch mode analyzes the slide center; grid mode extracts overlapping patches for future multi-patch aggregation. Background-heavy patches can be filtered using a tissue fraction threshold.

```
medtrustxai infer --modality pathology \
  --image slide.png --tile-mode center --patch-size 512
```

6 Evaluation Metrics

- VQA match: normalized yes/no and keyword overlap against reference_hint
- Token F1: word-level overlap between prediction and reference
- Faithfulness score: heatmap region vs mentioned anatomy/tissue terms
- CLI: medtrustxai evaluate --modality radiology

7 Usage

```
pip install -e ".[ui]"
python scripts/download_test_dataset.py --modality all
medtrustxai batch --modality radiology --no-xai
medtrustxai batch --modality pathology --no-xai
medtrustxai evaluate --modality radiology
medtrustxai serve --port 7860
```

8 Future Work

- Multi-patch WSI aggregation with attention-weighted fusion
- Clinical text + imaging multi-modal fusion
- Expert radiologist/pathologist review studies
- Larger benchmarks: MIMIC-CXR, Camelyon, PatchCamelyon

9 Conclusion

MedTrustXAI delivers a working multi-modal explainable AI prototype aligned with the research theme of trustworthy AI-assisted diagnosis in radiology and digital pathology. Both modalities are implemented with shared VLM and XAI infrastructure, modality-specific trust evaluation, WSI tiling, and quantitative metrics suitable for academic demonstration.